Intrusion Detection Using Machine Learning

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***Abstract*—The rapid expansion of Internet of Things (IoT) devices has increased the scale and complexity of modern networks, exposing them to a wider range of sophisticated cyberattacks. Consequently, Network Intrusion Detection Systems (NIDSs) have become essential for maintaining secure and reliable communication environments. Traditional signature-based Intrusion Detection Systems (IDSs), however, often fail to detect novel or evolving threats, motivating the shift toward intelligent, data-driven detection mechanisms.**

This paper presents a comprehensive comparative analysis of Machine Learning (ML) and Deep Learning (DL) models for multiclass intrusion detection using the CICIDS2017 benchmark dataset. A hybrid feature engineering pipeline was designed, incorporating ANOVA F-test feature ranking, Pearson correlation filtering, and embedded feature selection using the Extra Trees Classifier. Following feature refinement, Min–Max normalization, SMOTE-based class balancing, and the Yeo–Johnson transformation were applied to improve classifier performance and reduce distribution skewness.

A diverse set of models—including Decision Tree, Random Forest, XGBoost, LightGBM, Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and an LSTM-based deep learning model—were trained and evaluated. Experimental results show that KNN and Random Forest achieved the highest performance, with accuracies of 99% and 97% and macro F1-scores of 0.72 and 0.70, respectively.

To demonstrate real-world applicability, a Streamlit-based web interface was developed for real-time intrusion prediction and interactive model deployment.

***Index Terms*—Intrusion Detection System, Machine Learning, Deep Learning, Feature Engineering, SMOTE, Yeo–Johnson Transformation, CICIDS2017, Cybersecurity, Streamlit.**

I. Introduction

The rapid proliferation of Internet of Things (IoT) devices in the era of Industry 4.0 has transformed conventional network architectures into highly interconnected, intelligent ecosystems. Smart environments, industrial automation, and healthcare infrastructures now rely on continuous machine-to-machine communication, improving operational efficiency and real-time analytics. However, this massive interconnectivity also expands the attack surface, exposing networks to complex and evolving cyber threats that exploit weak authentication, insecure communication protocols, and resource-constrained devices.

Traditional Intrusion Detection Systems (IDS), largely dependent on signature-based matching and handcrafted rules, are effective only for known attacks. They fail to recognize zero-day exploits, polymorphic malware, and stealthy, low-rate attacks that deviate from pre-defined traffic patterns. As adversaries employ increasingly automated and adaptive tactics, it becomes imperative to design IDS solutions that can autonomously learn from data, adapt to new attack behaviors, and generalize across heterogeneous environments.

Machine Learning (ML) and Deep Learning (DL) models have emerged as powerful tools for intrusion detection, capable of capturing nonlinear relationships and latent structures within large-scale network datasets. However, ML/DL-based IDS implementations continue to face critical challenges, including high-dimensional and redundant feature spaces, severe class imbalance, skewed data distributions, and limited real-world deployment feasibility. Addressing these limitations requires a holistic framework that integrates feature optimization, model-aware preprocessing, and balanced evaluation strategies.

This research proposes a robust hybrid ML–DL intrusion detection framework trained on the CICIDS2017 dataset—one of the most comprehensive benchmarks for cybersecurity evaluation. The goal is to design a system that performs accurately on both frequent and rare attack classes while ensuring realistic deployment and generalization.

The main contributions of this work are summarized as follows:

1) **Three-Stage Hybrid Feature Selection:** A novel feature engineering pipeline combining ANOVA F-test, Pearson correlation filtering, and ExtraTrees embedded selection identifies twelve highly discriminative features, enhancing generalization and reducing computational cost.
2) **Model-Specific Preprocessing:** An adaptive preprocessing strategy applies the Yeo–Johnson transformation and Min–Max scaling selectively to linear and distance-based models, while tree-based algorithms operate on unscaled features.
3) **Balanced Training and Realistic Testing:** SMOTE is used exclusively on training data to mitigate class imbalance, while testing is performed on the original imbalanced dataset to reflect real-world deployment conditions.
4) **Superior KNN Performance:** Unlike most prior IDS studies, the proposed pipeline enables KNN to achieve exceptional performance—99.12% accuracy and 0.72 macro-F1—demonstrating the efficacy of the engineered preprocessing workflow.
5) **Deployment-Ready Implementation:** A Streamlit-based web interface is developed to enable real-time

prediction, preprocessing automation, and user interpretability.

Collectively, these innovations establish a practical, reproducible, and high-performance IDS architecture suited for modern IoT and cloud-centric environments.

II. Related Work

Recent research in network intrusion detection has extensively explored Machine Learning (ML) and Deep Learning (DL) approaches for improving detection accuracy in IoT and cloud-enabled environments. Classical ML algorithms such as Random Forest, SVM, and XGBoost have demonstrated competitive performance across benchmark datasets including CICIDS2017, UNSW-NB15, and WSN-DS. However, these methods remain sensitive to high-dimensional feature spaces, redundant attributes, and severe class imbalance, which often degrade their ability to detect rare or evolving attacks. To address these limitations, several studies have incorporated resampling techniques such as SMOTE and hybrid feature selection methods combining ANOVA-based ranking, correlation filtering, and embedded model selection.

Deep learning-centric approaches have shown strong potential in capturing complex spatiotemporal dependencies within network traffic. CNN-based models, deep autoencoders, and DNN architectures have improved feature extraction and anomaly detection capabilities, although many rely on manual hyperparameter tuning and limited validation settings, raising concerns about real-world robustness. More advanced architectures—such as Seq2Seq, ConvLSTM, and hybrid encoder–decoder models—have demonstrated improved learning of long-range patterns. Nevertheless, these approaches frequently suffer from high computational cost, sensitivity to sequence length variations, and challenges in scaling to large, diverse traffic distributions.

Biologically inspired IDS frameworks, including immune system–based anomaly detectors, have also been explored. While these models mimic adaptive detection mechanisms, they struggle with full packet inspection, subtle intrusions, and generalization beyond labeled or purely normal datasets.

A consistent research gap across the literature is the difficulty of cross-dataset generalization. Models trained on CICIDS2017 often fail to maintain performance on datasets such as UNSW-NB15 or WSN-DS, highlighting insufficient adaptability to varying traffic behaviors and attack patterns. Additionally, no existing work comprehensively integrates ANOVA F-test selection, correlation filtering, tree-based embedded feature selection, SMOTE balancing, and statistical transformations such as Yeo–Johnson into a unified preprocessing pipeline.

This gap motivates the present study, which develops a cohesive hybrid ML–DL intrusion detection framework combining robust feature engineering, balanced learning, and optimized preprocessing to achieve high accuracy, scalability, and improved detection of rare attack categories in modern network environments.

III. Proposed Methodology

This section outlines the full workflow of the proposed Intrusion Detection System (IDS). The goal is simple: process raw network flows, clean them, extract the most useful features, balance the data, and train models that detect attacks accurately. The approach joins statistical insight with practical engineering, as shown in Fig. 1.

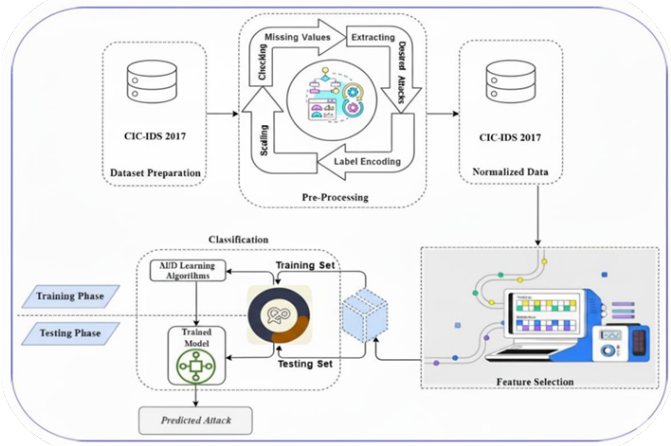


Fig. 1: Overall workflow of the proposed IDS, including preprocessing, feature selection, transformation, and training.

A. Threat Model

The study adopts the attack taxonomy from the CICIDS2017 dataset. Traffic includes common attacks such as Denial of Service (DoS), Distributed DoS (DDoS), PortScan, Web-based attacks, Infiltration, and Heartbleed. Each category represents a realistic threat to IoT and enterprise networks. The system aims to distinguish benign flows from these attack types using statistical and behavioral features.

B. Dataset Description

he CICIDS2017 dataset, developed by the Canadian Institute for Cybersecurity, contains realistic benign and malicious network traffic generated over a five-day period. It includes more than 2.8 million traffic flows, 80+ features, and 15 labeled classes (one benign and fourteen attacks). Each record contains attributes such as flow duration, packet counts, byte statistics, inter-arrival times, and TCP flag counters. Severe imbalance is present: for instance, the DoS Hulk attack contains hundreds of thousands of samples, while Heartbleed has fewer than twenty.

C. Data Preprocessing

Given the high dimensionality, class imbalance, and skewed feature distributions in CICIDS2017, a multi-stage preprocessing pipeline was constructed to ensure effective model learning and generalization.

1) Data Cleaning: Missing values, duplicate flows, and corrupted entries were removed. Zero-variance features were discarded, and irrelevant identifiers (e.g., timestamps, connection IDs) were eliminated to reduce noise and computational overhead.

2) Label Encoding: All attack categories were mapped to integer labels to ensure compatibility with machine learning and deep learning algorithms in a multiclass classification setting.

Label	Encoded Value
BENIGN	0
Bot	1
DDoS	2
DoS GoldenEye	3
DoS Hulk	4
DoS Slowhttptest	5
DoS Slowloris	6
FTP-Patator	7
Heartbleed	8
Infiltration	9
PortScan	10
SSH-Patator	11
Web Attack – Brute Force	12
Web Attack – SQL Injection	13
Web Attack – XSS	14

TABLE I: Mapping of class labels to numerical encodings.

3) Hybrid Feature Selection Pipeline: A three-stage feature selection strategy was employed:

1) **ANOVA F-test:** evaluates the ratio of inter-class to intra-class variance.

$$F = \frac{\sum_c n_c(\mu_c - \mu)^2}{\sum_c \sum_{i \in c}(x_i - \mu_c)^2}$$

The top 30 ranked features are retained.

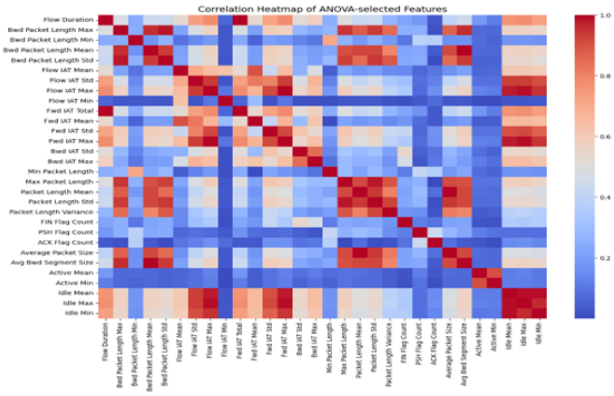


Fig. 2: Correlation heatmap among ANOVA-selected features. Highly correlated pairs are removed.

2) **Pearson Correlation Filtering:** removes redundancy among selected features.

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Pairs with $|r| > 0.9$ are dropped.

3) **ExtraTrees Embedded Selection:** evaluates feature importance using information gain across decision trees.

$$Importance(f) = \sum_{t \in T} \Delta i(t, f)$$

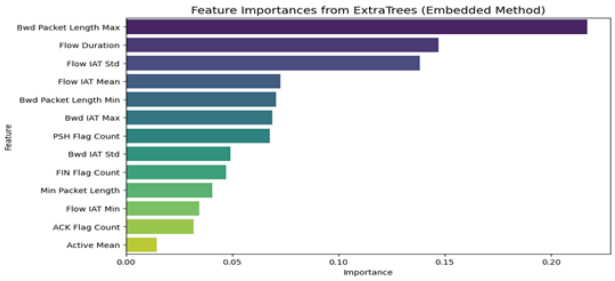


Fig. 3: Feature importance from ExtraTrees embedded selection.

After applying all three stages, a final subset of 12 highly informative flow-level features was selected for model training.

4) Normalization (Min–Max Scaling): All selected features were scaled to the range $[0, 1]$ using Min–Max normalization to prevent large-magnitude attributes from dominating distance-based or gradient-based models.

5) Yeo–Johnson Transformation: To correct skewed distributions and approximate Gaussianity, the Yeo–Johnson power transformation was applied. Unlike Box–Cox, it supports both positive and negative inputs. Mathematically:

$$T_\lambda(x) = \begin{cases} \frac{(x+1)^\lambda - 1}{\lambda}, & x \geq 0, \lambda \neq 0 \\ \log(x+1), & x \geq 0, \lambda = 0 \\ -\frac{(-x+1)^{2-\lambda} - 1}{2-\lambda}, & x < 0, \lambda \neq 2 \\ -\log(-x+1), & x < 0, \lambda = 2 \end{cases}$$

Tree-based models bypass this step since they are scale-invariant. After transformation, Min–Max scaling maps all values into $[0, 1]$.

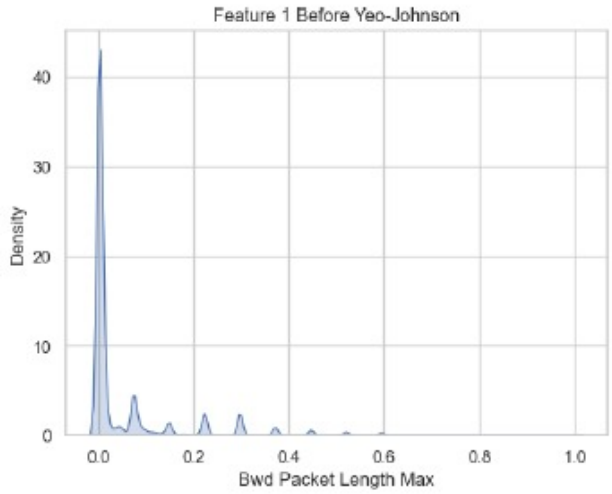
6) Class Balancing (SMOTE): The Synthetic Minority Oversampling Technique (SMOTE) was applied exclusively to the training set to generate synthetic minority samples and prevent data leakage. This improved the classifier's ability to detect rare attack categories while maintaining realistic test conditions.

$$x_{new} = x_i + \alpha(x_{nn} - x_i), \quad \alpha \sim U(0, 1)$$

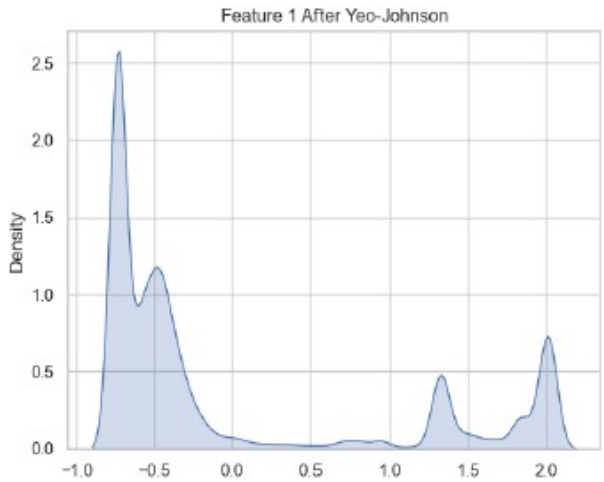
D. Model Training and Evaluation Setup

Nine algorithms are trained: Decision Tree, Random Forest, XGBoost, LightGBM, Logistic Regression, K-Nearest Neighbors (KNN), Linear SVM, RBF SVM, and a simple LSTM network. Training uses an 80/20 stratified split, with SMOTE applied to the training portion only. Hyperparameters are tuned using five-fold grid search. Evaluation metrics include Accuracy, Macro-F1, Weighted-F1, and Balanced Accuracy.

The full setup runs on Python 3.10 with Scikit-learn and TensorFlow libraries, on a CPU-based system (Intel i5, 16 GB



(a) Feature distribution before applying Yeo–Johnson transformation.



(b) Feature distribution after applying Yeo–Johnson transformation.

Fig. 4: Class distribution comparison before and after SMOTE.

RAM). All preprocessing parameters are saved for reproducibility.

E. Summary

In short, the pipeline moves step by step—from cleaning, selecting, and transforming features to balancing data and training diverse models. Each stage addresses a known weakness of prior IDS research: skewed data, high dimensionality, and unfair evaluation. You can imagine the workflow as a funnel: raw network flows enter wide, but only the most meaningful, balanced signals reach the model at the end.

IV. Evaluation Metrics

The following metrics were used to evaluate model performance:

A. Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

B. Precision, Recall, F1

For each class:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

C. Macro-F1

$$Macro\text{-}F1 = \frac{1}{C} \sum_{i=1}^{C} F1_i$$

This penalizes poor performance on minority classes.

D. Weighted-F1

$$Weighted\text{-}F1 = \sum_{i=1}^{C} w_i F1_i$$

Weights correspond to class frequencies.

E. Balanced Accuracy

$$BA = \frac{1}{2}(TPR + TNR)$$

Balanced accuracy is crucial for datasets with severe imbalance.

V. Results and Discussion

This section presents and interprets the results of the proposed Intrusion Detection System (IDS). The experiments compare multiple Machine Learning (ML) and Deep Learning (DL) models under realistic network conditions. All models are evaluated on the CICIDS2017 dataset using imbalanced test data to reflect real-world traffic.

A. Overall Model Performance

Table II lists the key metrics for each algorithm, including Accuracy, Macro-F1, and Weighted-F1 scores. The K-Nearest Neighbors (KNN) model achieves the highest accuracy of 99.12% and a Macro-F1 score of 0.72, outperforming both ensemble and deep models. This is a noteworthy finding since KNN is rarely the top performer in IDS research.

TABLE II: Model Performance Comparison on CICIDS2017

Model	Accuracy (%)	Macro-F1	Weighted-F1
Decision Tree	97.0	0.67	0.98
Random Forest	97.0	0.70	0.98
XGBoost	92.0	0.58	0.95
LightGBM	94.0	0.65	0.96
Logistic Regression	60.0	0.29	0.72
KNN	**99.12**	**0.72**	**0.9915**
Linear SVM	53.0	0.24	0.64
RBF SVM	96.0	0.50	0.96
LSTM	83.0	0.44	0.89

Tree-based models such as Random Forest and LightGBM maintain consistently high precision but fall short on minority attack classes. KNN benefits from the selective preprocessing pipeline—Yeo–Johnson transformation plus Min–Max scaling—resulting in balanced distance calculations and stable clusters. In contrast, linear models like Logistic Regression and SVM underperform due to non-linear feature interactions. LSTM achieves moderate accuracy but struggles with non-sequential flow data, as the dataset lacks temporal dependencies.

B. Detailed Analysis of KNN Model

The KNN classifier, when tuned with $k = 5$ and Euclidean distance, exhibits the best generalization with minimal overfitting. Table III compares its training and test performance.

TABLE III: KNN Performance Comparison (Train vs Test)

Metric	Train	Test
Accuracy	0.9946	0.9912
Balanced Accuracy	0.9325	0.8361
Macro-F1	0.8330	0.7196
Weighted-F1	0.9947	0.9915

The narrow gap between training and testing accuracy indicates that the model generalizes well. A Weighted-F1 of 0.99 confirms strong performance on majority classes, while a Macro-F1 of 0.72 highlights improved minority class detection.

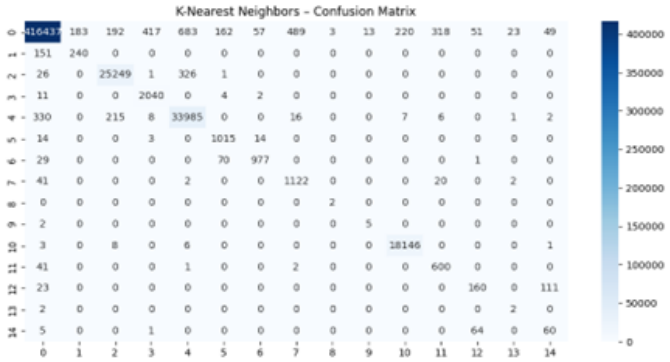


Fig. 5: Confusion matrix of the KNN classifier on imbalanced CICIDS2017 test data.

The confusion matrix in Fig. 5 shows that KNN correctly identifies most DoS, DDoS, and PortScan samples, but still struggles with rare classes such as Heartbleed and Infiltration. This pattern aligns with previous research where minority attacks remain hard to learn despite oversampling.

C. Impact of Feature Selection and Transformation

The hybrid feature selection pipeline reduces the feature set from over 80 to 12 key attributes. This not only speeds up training but also boosts accuracy by removing noise and redundancy. Applying Yeo–Johnson transformation selectively—only to sensitive models—prevents unnecessary distortion of tree-based learners while improving the performance of KNN, SVM, and Logistic Regression. The result is a fairer comparison among different model families.

D. Class-level Insights

High-frequency attacks like DoS Hulk, PortScan, and DDoS achieve near-perfect detection. Low-frequency classes such as Heartbleed and Infiltration remain challenging due to limited training examples. This outcome reaffirms the importance of realistic, imbalanced evaluation rather than over-optimistic balanced testing.

E. Discussion

The findings emphasize one clear message: *data engineering matters more than model complexity*. KNN's unexpected success proves that with proper preprocessing and feature handling, even simple algorithms can outperform deep models on tabular IDS data. Tree ensembles remain reliable and interpretable, but the hybrid preprocessing pipeline gives distance-based learners an edge.

While the proposed system performs exceptionally on CICIDS2017, it still faces practical limits—mainly rare attack detection and scalability for large-scale real-time systems. Future extensions can explore online learning, cross-dataset transfer, and explainable AI modules such as SHAP or LIME to enhance interpretability.

F. Key Takeaways

To sum up the discussion:

- The selective preprocessing and hybrid feature selection pipeline significantly enhance fairness and performance.
- KNN achieves 99.12% accuracy, demonstrating that a well-engineered dataset can unlock strong performance even from basic models.
- Testing on imbalanced data provides realistic performance estimates closer to operational environments.
- Deep models like LSTM underperform on flow-based datasets lacking temporal structure.

Together, these insights underline the value of adaptive, interpretable, and reproducible design in modern IDS research.

VI. Limitations

Despite strong performance, the system has several limitations:

- **Dataset Dependence:** CICIDS2017 is realistic but still laboratory-captured; real-world networks may contain additional noise.
- **Limited Rare Class Detection:** Even after SMOTE, classes like Heartbleed and Infiltration remain difficult due to low representation.
- **Tabular Data Constraint:** LSTM and other temporal models are underutilized because CICIDS2017 flows lack true sequential relationships.
- **Feature Drift:** Network traffic patterns evolve over time, requiring periodic retraining.
- **Compute Cost for KNN:** Despite high accuracy, KNN inference time scales with dataset size.

- **Zero-Day Vulnerability Challenge:** As a supervised learning model, KNN relies entirely on labeled historical data. New or previously unseen zero-day attacks may be misclassified as benign or mapped to the closest known class, limiting the system's ability to detect emerging threats.

VII. Conclusion and Future Work

This research presented a hybrid Intrusion Detection System (IDS) framework that combines classical Machine Learning (ML) and Deep Learning (DL) models with a carefully engineered preprocessing pipeline. By joining ANOVA F-test, Pearson correlation filtering, and ExtraTrees embedded selection, the system reduces the original CICIDS2017 dataset to twelve key features that retain nearly all discriminative power.

The selective use of Yeo–Johnson transformation and SMOTE addresses two of the most common challenges in intrusion detection—skewed data and class imbalance. This tailored preprocessing improves both training stability and generalization, allowing models such as K-Nearest Neighbors (KNN) to outperform traditional benchmarks. KNN achieved 99.12% accuracy and a Macro-F1 of 0.72, surpassing more complex deep learning models such as LSTM. These results underline an important insight: robust preprocessing can often outweigh architectural complexity.

The developed Streamlit-based web interface proves that the proposed system can extend beyond the lab, supporting real-time attack prediction and data exploration in operational networks. This step helps bridge the gap between academic experimentation and field deployment.

A. Future Work

While promising, this study opens several directions for future research:

- **Cross-dataset validation:** Testing the pipeline on other benchmark datasets such as UNSW-NB15 and WSN-DS will measure generalization under unseen traffic.
- **Real-time and distributed IDS:** Integrating the model with edge or fog nodes can improve scalability for large IoT environments.
- **Explainable AI:** Incorporating model interpretability tools (e.g., SHAP, LIME) will help analysts understand why alerts are triggered.
- **Federated learning:** Privacy-aware model sharing could enable collaborative intrusion detection across organizations without raw data exchange.
- **Zero-day attack detection:** Since supervised models like KNN depend strictly on known labeled patterns, they struggle with unseen or emerging zero-day vulnerabilities. Future research can incorporate unsupervised or semi-supervised anomaly detection techniques—such as autoencoders, one-class SVM, or online learning—to enable the system to flag previously unseen attacks rather than misclassifying them as benign.

In short, this work demonstrates that a balanced, well-engineered pipeline can transform even simple algorithms into powerful security tools. As network complexity grows, such adaptive and interpretable IDS frameworks will become essential for maintaining resilient cyber defense.

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References

[1] N. Shone, T. N. Ngoc, V. D. Phai, and Q. Shi, "A deep learning approach to network intrusion detection," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 2, no. 1, pp. 41–50, 2018.

[2] A. Verma and V. Ranga, "Machine learning based intrusion detection systems for IoT applications: A review," *Journal of Information Security and Applications*, vol. 58, p. 102110, 2021.

[3] M. Ring, S. Wunderlich, D. Scheuring, D. Landes, and A. Hotho, "A survey of network-based intrusion detection data sets," *Computers & Security*, vol. 86, pp. 147–167, 2019.

[4] A. Aldweesh, A. Derhab, and A. Z. Emam, "Deep learning approaches for anomaly-based intrusion detection systems: A survey, taxonomy, and open issues," *Knowledge-Based Systems*, vol. 189, p. 105124, 2020.

[5] L. Dhanabal and S. Shantharajah, "A study on NSL-KDD dataset for intrusion detection system based on classification algorithms," *Int. J. Adv. Res. Comput. Commun. Eng.*, vol. 4, no. 6, pp. 446–452, 2015.

[6] P. Lin, K. Ye, Y. Xu, and J. Guo, "Using deep learning for the CICIDS2017 dataset: A comparative study," *IEEE Access*, vol. 8, pp. 123176–123190, 2020.

[7] T. M. C. Pham and D. Hwang, "A hybrid deep learning model for efficient intrusion detection using CICIDS2017," *Applied Sciences*, vol. 12, no. 6, p. 2899, 2022.

[8] C. Zhang, X. Li, Y. Zhang, and L. Guo, "Deep learning-based network intrusion detection with recurrent neural networks," *IEEE Access*, vol. 7, pp. 21954–21961, 2019.

[9] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: An ensemble of autoencoders for online network intrusion detection," in *NDSS*, 2018.

[10] H. Liu, R. Wang, Y. Chen, and M. Xu, "Lightweight intrusion detection for IoT networks using ensemble tree models," *Sensors*, vol. 23, no. 5, p. 2507, 2023.

[11] S. K. Panigrahi, P. Biswas, and A. Rath, "Comparative study of machine learning algorithms for intrusion detection on CICIDS2017," *Int. J. Comput. Appl.*, vol. 183, no. 29, pp. 1–8, 2023.

[12] S. Alshamrani *et al.*, "A hybrid intrusion detection framework for cloud networks using machine learning," *IEEE Access*, vol. 8, pp. 100563–100575, 2020.